SHREYAS KRISHNAREDDY

AI Engineer | Full-Stack Software Engineer

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PROFILE SUMMARY

AI Engineer and Full-Stack Software Engineer with hands-on experience building production-grade intelligent systems. Core expertise in scalable microservices, high-performance APIs, and end-to-end AI pipelines using Python, FastAPI, PostgreSQL, and Docker on AWS. Deep hands-on experience with LLM integration (GPT, Claude, Llama, Mistral), RAG architectures, semantic vector search, prompt engineering, and autonomous AI agent systems including MCP-based agent orchestration.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS, Bash

Frameworks & Tools: FastAPI, React, Next.js, Node.js, Flask, Docker, Kubernetes, AWS (ECS, S3), Azure, Git, CI/CD, Linux

AI/ML & Agents: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LlamaIndex, OpenAI SDK, MCP, RAG, Fine-Tuning, Prompt Engineering

NLP & Search: Embeddings, Semantic Search, Cosine Similarity, Vector Databases (Pinecone, FAISS, Weaviate, Chroma DB), NER, Text Extraction, Reranking

Databases: PostgreSQL, Redis, MongoDB, Neo4j, Supabase, SQLite

MLOps & Observability: Docker, Kubernetes, CI/CD, Model Versioning, Model Monitoring, Production Logging, Performance Profiling

EDUCATION

Texas A&M University – Corpus Christi — *M.S. Computer Science* | Corpus Christi, TX | 08/2023 – 05/2025

Bharathiar University — *M.S. Data Analytics* | Coimbatore, India | 06/2019 – 05/2021

WORK EXPERIENCE

Arytic, Inc | Renton, WA

AI Engineer | Aug 2025 – Present

- Reduced manual document review time by 85% by architecting an AI-powered document validation platform that processes 10,000+ documents across 42 data fields at 99.7% accuracy using a scalable microservices pipeline built with Python, FastAPI, and Docker.
- Designed and built a candidate-first semantic matching pipeline that processes full resume text through PDF text extraction, embedding generation, and vector search using cosine similarity to deliver ranked job matches with transparent match percentage scores across 83,000+ active positions.
- Engineered multi-source job aggregation from ATS platforms (Greenhouse, Lever, Ashby, Workday) with automated daily refresh cycles, query caching for sub-second response times, and IP-based rate limiting.
- Eliminated vendor lock-in by designing a pluggable LLM provider architecture supporting GPT, Llama, Groq, and Ollama with configuration-driven backend switching, cutting model experimentation time from days to minutes.
- Built high-performance asynchronous REST APIs using FastAPI, PostgreSQL, and Redis, containerized with Docker, serving concurrent document processing requests with horizontal scaling.
- Implemented a Model Context Protocol (MCP) server exposing platform capabilities as programmable tools, enabling seamless third-party AI agent integration and interoperability with external agent frameworks.
- Reduced LLM API costs by 40% through prompt optimization across GPT and open-source models, balancing extraction accuracy, token usage, and inference latency.

Giving Forward | Remote

Software Engineer | Jul 2025 – Aug 2025

- Built scalable data pipelines aggregating financial and operational metrics across 62+ nonprofit organizations ($425M+ AUM), replacing manual spreadsheet workflows with real-time REST API access using Python, FastAPI, and PostgreSQL.
- Shipped semantic search using TF-IDF vectorization and LLM-based reranking over organizational data, reducing average query-to-insight time from minutes to under 2 seconds.
- Designed RESTful API endpoints with data preprocessing, caching strategies (Redis, PostgreSQL), and feature engineering to deliver high-performance analytics dashboards.

Texas A&M University | Corpus Christi, TX
Research Assistant — NLP & Knowledge Graphs | Jan 2024 – May 2025

- Built a multi-hop retrieval system over Neo4j knowledge graphs using graph-based reasoning and Retrieval-Augmented Generation (RAG) that outperformed single-hop baselines on 3 benchmark datasets.
- Designed graph traversal algorithms and embedding-based semantic search using vector embeddings to enable contextual, multi-step reasoning across interconnected knowledge bases.
- Trained and fine-tuned language models with PyTorch and Hugging Face Transformers on domain-specific QA datasets, improving retrieval accuracy and answer quality.
- Automated end-to-end experiment evaluation pipelines with reproducible metric computation (Precision, Recall, F1, Accuracy), reducing analysis cycles from hours to minutes.
- Published research on multi-hop reasoning over knowledge graphs, contributing to graph-augmented RAG architectures.

PROJECTS

HireMe
Autonomous Job Application Agent

- Built an autonomous AI agent that discovers, evaluates, and applies to jobs end-to-end using Claude Code as a subprocess with MCP-based browser control (Playwright), adapting to any ATS form without pre-built selectors.
- Engineered multi-worker parallel execution with isolated Chrome instances, atomic SQLite job acquisition (WAL mode), and thread-safe concurrency across 4,800 lines of production Python.
- Implemented CAPTCHA detection and solving pipeline (hCaptcha, reCAPTCHA, Cloudflare Turnstile), intelligent failure classification with retry logic, and real-time monitoring dashboards (Rich + Streamlit).

Deep Research Agent
Constraint-Bounded Multi-Agent Research System

- Built a multi-agent research system that decomposes complex queries, searches the web, and synthesizes structured answers under strict memory (4K token) and cost ($0.10/session) budgets using a three-tier summarization cascade.
- Implemented dual-model cost optimization (GPT-4o-mini for extraction, reasoning model for synthesis), Docker Compose deployment, and provably bounded resource constraints with exact token tracking.

AI-Powered Assistant
Real-Time Video and Audio Processing

- Designed and deployed FastAPI endpoints for a real-time multimodal processing system integrating OpenCV, Whisper (STT/TTS), and GPT-4, achieving sub-500ms latency for gesture recognition and speech-to-text conversion to enable seamless human-AI interaction.

CERTIFICATIONS

- Prompt Design in Vertex AI, *Google Cloud*
- Knowledge Graphs for RAG, *DeepLearning.AI*
- Finetuning Large Language Models, *DeepLearning.AI*
- AI Agents Fundamentals, *Hugging Face*